



Servlet III

1. Attributes and listeners
2. State management
3. Handling cookies

- Attributes are objects which are used to share information between multiple servlets and JSP.
- Attributes can be set or get from 3 different scopes
 - Request: Request attributes end when the response is sent back to client.
 - Session: Session level attributes will be valid till user session is active.
 - Application: This scope persists until the application is deployed.
- Methods used:
 - `public void setAttribute(String name, Object object)`
 - `public Object getAttribute(String name)`
 - `public void removeAttribute(String name)`

- Listeners
 - Servlet Listener is used for listening to events in web containers, such as when you create a session, insert an attribute.
 - There are total eight type of listeners available in servlet framework
 - Theses listeners are interfaces which needs to be implemented by classes and these classes should be annotated with @WebListener annotation.

- Need of state management:.
 - HTTP is a stateless (or non-persistent) protocol
 - Each request is considered as fresh request.
 - Sometimes it is needed to keep track of client's activity across multiple requests.
- What is state management
 - "State" is any data that describes the current behavior of an application.
 - State management is nothing but saving data on the server or client which might be required in future for processing further requests.
 - State management takes help of different techniques to save and update the required information

- State management techniques
 - There are several techniques used by Servlet application for state management.
 - i. Hidden form field
 - ii. URL Rewriting
 - iii. Cookies
 - iv. HttpSession

- A cookie is a small amount of data which is stored in the web browser.
- Transferred between requests and responses through HTTP headers.
- A cookie has a name and value, plus few optional attributes
- Pros: Can be used without server resources.
- Cons: Increases network traffic, security risks, size limitation
- Creating cookie on server side
 - Create cookie object, set required properties of cookie object and send it in the response
- Retrieving cookie from request
 - Request object provides method `getCookies()` which returns array of cookies

- This array can be iterated to find the required cookie by using name of the cookies.
- Updating cookie.
 - Create the cookie with same name and updated value and send it in the response
- Deleting cookie
 - Set the max age of the cookie as 0(zero) and send it in the response. Cookie will be deleted by the browser